

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50° C [120° F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

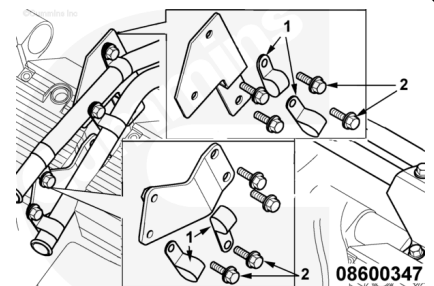
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/-008-018.html>).

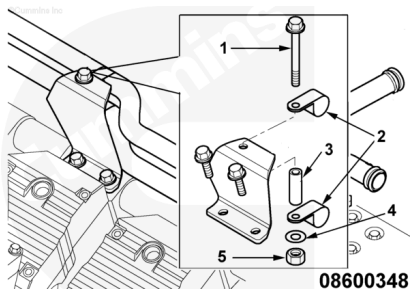
Remove

Right Bank

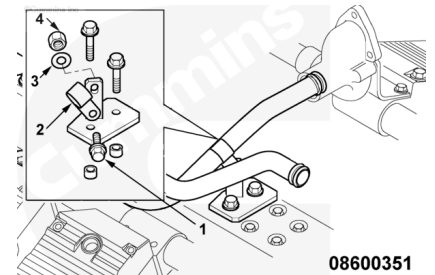
Remove the P-clips (1) and capscrews (2) which mount the upper and lower tubes to the brackets on the number 5 rocker housing.



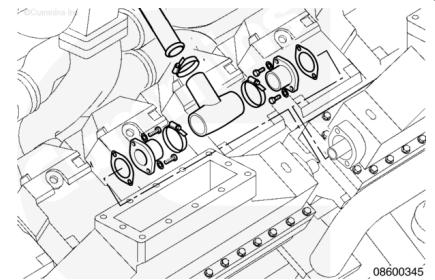
Remove the capscrew (1), P-clips (2), spacer (3), washer (4), and nut (5) that mount both the upper and lower water tubes to the support bracket, mounted on rocker housing number 2.



Remove the capscrew (1), P-clip (2), washer (3), and nut (4) that mount the lower water tube to the bracket, mounted on thermostat housing support.



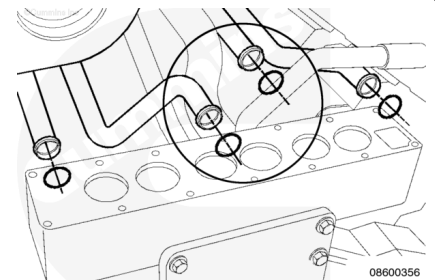
Remove the lower and upper water transfer tube from the rubber tee pieces.



Remove the upper water transfer tube from the thermostat housing.

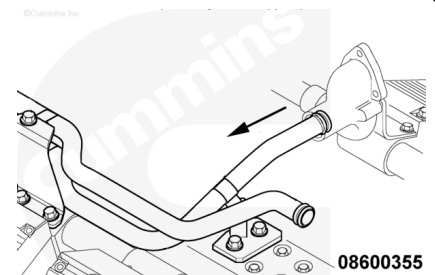
Remove and discard the o-rings.

Note : The circled area of the graphic is for location purposes only.



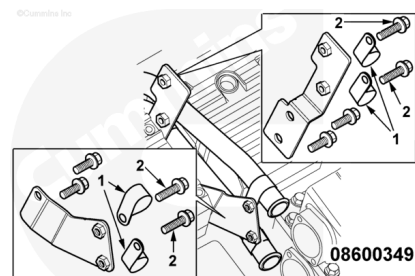
Remove the lower tube from the junction block.

Remove and discard the o-rings.

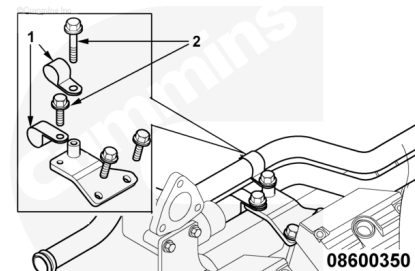


Left Bank

Remove the P-clips (1) and capscrews (2) that mount the upper and lower water transfer tubes to the brackets on rocker housing number 5.



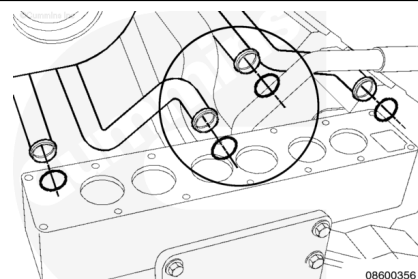
Remove the P-clips (1) and capscrews (2) that mount both the upper and lower water transfer tubes to the bracket mounted on rocker housing number 2.



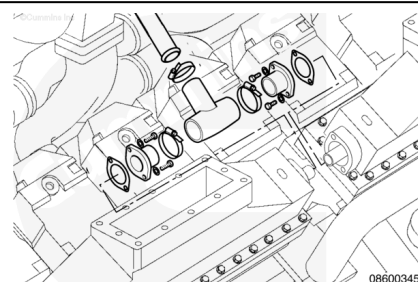
Remove the upper water transfer tube from the thermostat housing.

Remove and discard the o-rings.

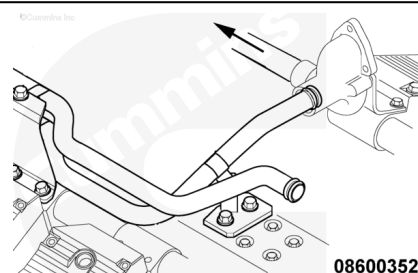
Note : The circled area of the graphic is for location purposes only.



Remove the upper and lower water tubes from the rubber tee pieces.

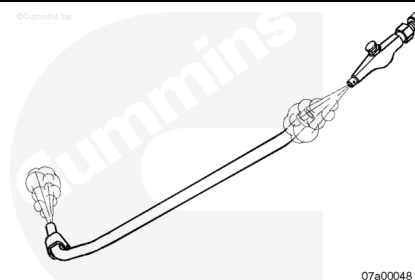


Remove the lower tube from the junction block.



Clean

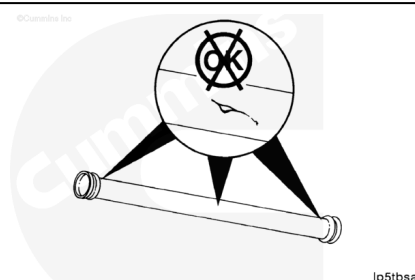
Flush the pipes with water to be sure the water pipes are internally clean and free of debris before re-fitting.



Inspect for Reuse

Check that the pipes are free from dents, cracks, etc.

Note : O-rings must always be replaced following removal of the water transfer tubes.



Install

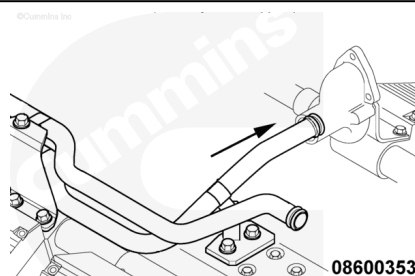
Right Bank

⚠ CAUTION ⚠

Do not use petroleum based oil. This will cause swelling of the o-ring.

Use vegetable oil to lubricate the o-ring.

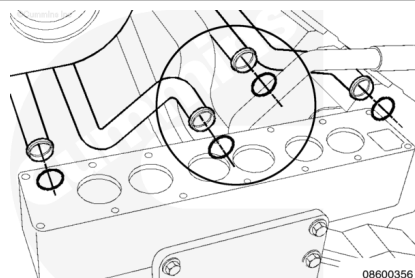
Fit the lower water tube into the junction block.



⚠ CAUTION ⚠

Do not use petroleum based oil. This will cause swelling of the o-ring.

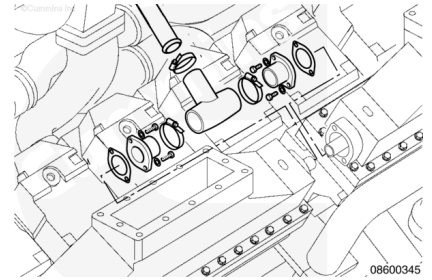
Use vegetable oil to lubricate the o-ring.



Fit the upper tube into the thermostat housing.

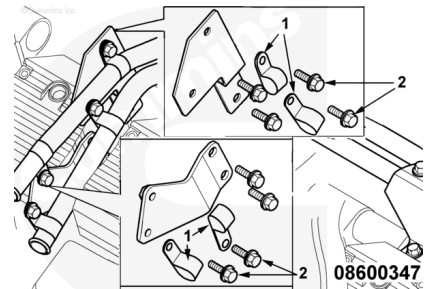
Note : The circled area of the graphic is for location purposes only.

Fit the upper and lower water tubes into the rubber tee pieces.



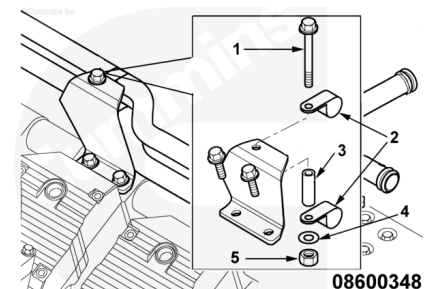
Secure both the upper and lower pipes to the brackets on the number 5 rocker housing using P-clips (1) and capscrews (2). To aid assembly, do **not** fully tighten the P-clip capscrews until the pipe is loosely clipped on all brackets.

Torque Value: 45 n•m [33 ft-lb]



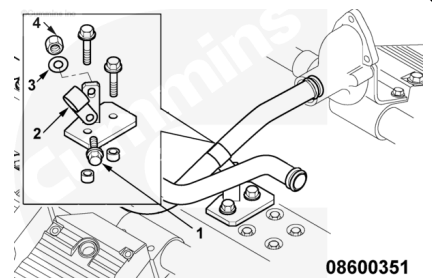
Secure both the upper and lower pipes to the brackets on the number 2 rocker housing using capscrews (1), P-clips (2), spacer (3) and nut (4). To aid assembly, do **not** fully tighten the P-clip capscrew until the pipe is loosely clipped on all the brackets.

Torque Value: 45 n•m [33 ft-lb]



Secure the lower water pipe to the bracket on the thermostat housing support.

Torque Value: 45 n•m [33 ft-lb]

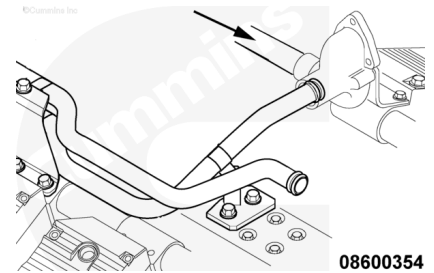


Left Bank

⚠ CAUTION ⚠

Do not use petroleum based oil. This will cause swelling of the o-ring.

Use vegetable oil to lubricate the o-ring; fit lower tube into the junction block.



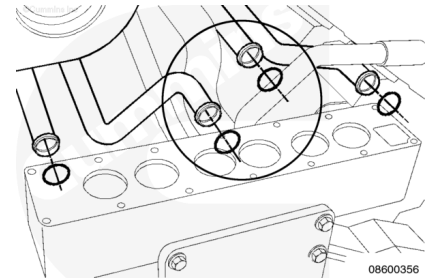
⚠ CAUTION ⚠

Do not use petroleum based oil. This will cause swelling of the o-ring.

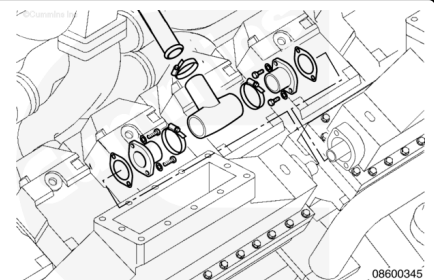
Use vegetable oil to lubricate the o-ring.

Fit upper tube into the thermostat housing.

Note : The circled area of the graphic is for location purposes only.

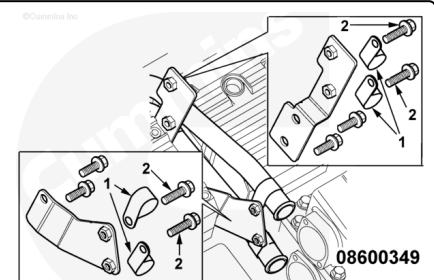


Fit the upper and lower water tubes into the rubber tee pieces.



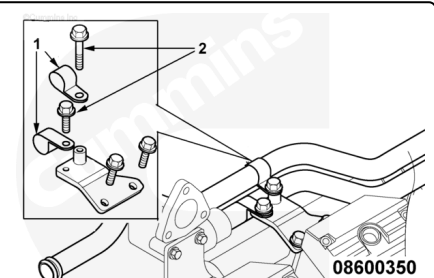
Secure both the upper and lower pipes to the brackets on the number 5 rocker housing using P-clips (1) and capscrews (2). To aid assembly, do **not** fully tighten the P-clip capscrews until the pipe is loosely clipped on all of the brackets.

Torque Value: 45 n•m [33 ft-lb]



Secure both the upper and lower pipes to the brackets on number 2 rocker housing using P-clips (1) and capscrews (2).

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

- Fill the cooling system. Refer to Procedure 008-018.
(/qs3/pubsys2/xml/en/procedures/-010-006.html)

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